ASU Micro Qual Module 3
Static Games, Extensive-Form Games, and Repeated/OLG Games

PhD Qual Lecture Notes Builder Packet

Overleaf-ready source

Field	ASU Microeconomic Theory Qualifier
Module	3 of 6: game theory core
Depth	Exhaustive but exam-facing
Target weakness	definitions, equilibrium theorems, graphical tools, computation, proof writing
Main answer format	Object → Feasibility → Equilibrium/Optimality → Verification
Suggested study time	three 3-hour blocks plus two old-qual timed drills

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1 Orientation: what this packet is built to do

Game-theory questions on the ASU Micro qualifier are usually not won by remembering a named result. They are won by writing the right object. A pure strategy in an extensive-form game is not an action; it is a complete contingent plan. A Nash equilibrium in a mixed extension is not just indifference on the support; it also requires off-support inequalities. A subgame-perfect Nash equilibrium is not merely a backward-induction outcome; it is a strategy profile that induces a Nash equilibrium in every subgame. A sequential equilibrium is not merely a strategy profile; it is an assessment, namely strategies plus beliefs, satisfying sequential rationality and consistency.

This packet therefore emphasizes five exam behaviors:

1. Define the game object before solving: players, strategy sets, histories/information sets, payoff representation.
2. Compute equilibrium with explicit best-response inequalities.
3. Separate path behavior from full strategy profiles.
4. Verify equilibrium at every relevant information set or subgame, not only on the equilibrium path.
5. Write compact proofs using reusable templates.

Four-layer answer. For any game problem, write:

1. **Object.** Define players, actions or histories, strategy sets, payoff functions, beliefs if needed.
2. **Feasibility.** Check strategies are complete contingent plans; probabilities sum to one; beliefs are valid and Bayes-consistent on path.
3. **Equilibrium/optimality.** State best-response, sequential-rationality, or incentive constraints.
4. **Verification.** Check every player, every support action, every off-support action, and every subgame/information set requested.

2 Source inventory

The source hierarchy used here is: past ASU qualifying exams first, current problem sets second, professor lecture notes/slides third, and supplementary textbook-level theory only to fill definitional gaps. Some older exam PDFs are scanned; where OCR was unreliable, the packet uses the visible exam page content and the legible extracted text.

Source	Type	Topics used	Extraction notes / exam relevance
GitHub folder `Microeconomics/Static Games`	Source folder	Folder inventory: normal form, extensive form, strategic-form examples, repeated prisoners' dilemma, repeated-games definitions, folk theorem, supermodular games.	Folder listing accessible; several PDFs also supplied locally. Non-uploaded raw PDFs were inventoried but not deeply extracted.
`MICRO_MANIFEST.md` and `SKILL.md`	Manifest and skill	Required Module 3 scope, four-layer answer style, source hierarchy, module-map/exercise requirements.	Directly governs packet structure.
`dgt01-eform.pdf`	Slide deck	Kuhn-style extensive form, game tree, terminal nodes, information sets, choices, chance moves, payoff map, perfect recall, perfect information, pure/mixed/behavior strategies, Kuhn equivalence.	Clean text extraction. Core definitions.
`dgt02-nform.pdf`	Slide deck	Normal-form representation of extensive-form games, strategy functions, expected-payoff computation, reduced normal form, agent-normal form.	Clean text extraction; used for complete-contingent-plan emphasis.
`NE-Existence.pdf`	Lecture note	Debreu–Fan–Glicksberg theorem, Nash mixed-existence corollary, mixed-support fact, affine programming proof.	Clean text extraction; central existence theorem source.
`SPMGames22.pdf`	Lecture note	Strictly supermodular games, Tarski fixed point theorem, largest/smallest Nash equilibria, monotone comparative statics.	Used as static-game theorem extension.
`RepeatedGames.pdf`	Lecture note	Histories H_t, strategies $f_i^t : H_{t-1} \to A_i$, discounted utility, minimax value.	Clean text extraction; used for repeated-game definitions.
`RepeatedGames FT.pdf`	Lecture note	Folk theorem with Nash reversion: target action payoff strictly above a stage-game NE payoff, sufficiently patient players.	Clean extraction; used for IC template.

Source	Type	Topics used	Extraction notes / exam relevance
`Class19--Class25(F25).pdf`	Class summaries	Strategic-form introduction; dominance, IDSDS, Nash; Hotelling; DFG; Cournot/Bertrand; extensive form and SPNE; repeated prisoners' dilemma.	Clean extraction. Professor-specific emphasis and notation.
`PS9(24)-2.pdf`	Problem set	IDSDS proof; mixed Battle of the Sexes; all-pay/product-development mixed equilibrium; DFG assumption checks.	Used for exercise bank and proof drills.
`PS10(25)-1.pdf`	Problem set	Extensive-form strategy sets; vertical integration; MWG 9.B.5; entry game with Bertrand/Cournot continuation; Nash vs SPNE.	Used for extensive-form practice and traps.
`PS11(25) (1).pdf`	Problem set	Stackelberg welfare; repeated MWG 9.B.9; finitely repeated prisoners' dilemma dominance claim; repeated Bertrand/collusion.	Used for repeated-game drills.
June 2019 qual	Past qual	Pure NE in a 3×3 game; twice-, three-, and four-times repeated game outcomes; advertising game and core.	High-priority Module 3 anchor.
June 2023 qual	Past qual	Owner-manager two-period game; strategy sets; highest/lowest SPNE owner payoff; Nash not SPNE; sequential equilibrium in an imperfect-information tree.	High-priority anchor for SPNE and sequential equilibrium.
August 2024 qual	Past qual	Stackelberg-Cournot extensive form; strategy sets; weak dominance; NE not SPNE; SPNE derivation.	High-priority anchor.
June 2024 qual	Past qual	Parametric mixed Nash equilibrium and equilibrium-correspondence continuity at $\beta = 1$.	Main support-enumeration anchor.

Source	Type	Topics used	Extraction notes / exam relevance
August 2021 qual	Past qual	OLG prisoners' dilemma; pure strategy sets; NE/SPNE support of young-cooperate/old-defect outcome; DFG non-applicability.	Main OLG anchor.
June 2018 qual	Past qual	Signaling game and three-person extensive-form game; pure/mixed NE; sequential equilibrium definition and verification.	Used for extensive/sequential practice; signaling details mainly deferred to Module 4.
June 2013 / August 2013 scanned quals	Past qual	Owner-manager predecessor problem; Selten horse sequential equilibrium; OLG prisoners' dilemma.	OCR limited; visible page content used.
June 2016 scanned qual	Past qual	Parent-child extensive-form game; strategy sets; SPNE claim; weak dominance.	OCR limited; used for practice map.

3 Detailed blueprint

3.1 Module objective

After this module, a student should be able to take any ASU game-theory question and, within five minutes, identify whether it is a static-form computation, an extensive-form strategy-set/SPNE problem, a sequential-equilibrium assessment problem, a repeated-game incentive-constraint problem, or an OLG-reputation problem. The student should then write the exact objects and inequalities needed for credit.

3.2 Prerequisites

- Basic constrained optimization and fixed-point logic.
- Expected utility notation and probability distributions.
- Familiarity with Cournot, Bertrand, monopoly output, and payoff matrices.
- Comfort with functions as strategies: $s_2 : A_1 \to A_2$ in a Stackelberg game.

3.3 Canonical models and exam prototypes

Canonical model	What the exam mutates	Core technique	Primary anchors
Strategic-form finite game	Pure NE, weak dominance, mixed NE, equilibrium-correspondence continuity.	Best-response tables; support enumeration; off-support checks.	June 2019 Q1; June 2024 Q2.
Continuous strategic game	Compactness/convexity, best-reply intersections, existence theorem applicability.	DFG assumptions; graphical best-response intersections; FOCs plus concavity.	Class20–22; NE-existence; PS9.
Stackelberg/leader-follower game	Strategy-set writing; strategic-form representation of an extensive form; weakly dominant follower strategies; NE not SPNE.	Complete contingent plans; backward induction; off-path punishment counterexamples.	August 2024 Q2; Class23; PS10.
Owner-manager/reputation game	Date-1 incentives with date-2 continuation values; highest/lowest SPNE payoff; NE not SPNE.	Continuation-value inequality $x+\delta \geq 1$; date-2 subgame verification.	June 2023 Q1; June 2013 Q1.
Sequential-equilibrium tree	Beliefs at an information set; sequential rationality; consistency; mixed strategies at ties.	Belief thresholds; Bayes rule; perturbation ratios.	June 2023 Q2; June 2018 Q4.
Finite repeated game	Last-period unraveling if unique stage NE; multiple stage NE can support non-stage actions early.	Backward induction; continuation rewards/punishments.	Class25; June 2019 Q1; PS11.
Infinite repeated game	Collusion/cooperation with trigger strategies; folk theorem intuition.	One-shot deviation IC: cooperate value vs deviation plus punishment.	RepeatedGames; RepeatedGames FT; Class25.
OLG repeated game	Strategy sets for finite-lived overlapping agents; NE/SPNE support; why DFG cannot be invoked blindly.	Young incentive constraint; old one-shot check; subgame continuation.	August 2021 Q1; August 2013 Q2.

3.4 Closed-form catalog

1. 2×2 **mixed equilibrium.** Set each player indifferent on support; then verify probabilities lie in $[0, 1]$ and off-support actions are not profitable.
2. **Linear Cournot with n symmetric firms.** $q_i^* = (a-k)/(n+1)$, $P^* = (a+nk)/(n+1)$, profit $((a-k)/(n+1))^2$.
3. **Bertrand homogeneous goods.** With identical constant unit cost k, unique pure NE price is $p_i = k$ under the textbook tie-breaking assumptions.

4. **Stackelberg-Cournot.** Follower response $R_2(q_1) = \max\{0, (a - k - q_1)/2\}$; leader $q_1^S = (a - k)/2$; follower $q_2^S = (a - k)/4$.
5. **Owner-manager best SPNE payoff.** Highest owner payoff is δ; choose date-1 expropriation $x = 1 - \delta$ and reward/punish with date-2 hiring.
6. **Finitely repeated prisoners' dilemma with unique stage NE.** Unique SPNE: defect after every history.
7. **Infinite repeated prisoners' dilemma with payoffs $T = 3, R = 2, P = 0, S = -1$.** Grim trigger supports cooperation iff $\delta \geq (T - R)/(T - P) = 1/3$.
8. **Repeated Bertrand collusion.** If deviation captures one-period monopoly profit and punishment yields zero thereafter, equal-split collusion needs $\delta \geq 1/2$.
9. **OLG young-C/old-D chain.** A young player cooperates iff $-1 + 3\delta \geq 0$, so $\delta \geq 1/3$.

3.5 Study plan

- **Block 1: static games.** Definitions, dominance, pure/mixed NE, DFG/Nash existence, Cournot/Bertrand, June 2024 Q2.
- **Block 2: extensive form.** Strategy-set writing, normal-form representation, SPNE, Stackelberg, owner-manager, June 2023 Q1 and August 2024 Q2.
- **Block 3: sequential and repeated.** Sequential equilibrium assessments, finite/infinite repeated games, folk theorem, OLG.
- **Timed drills.** Do June 2023 Q1–Q2 and August 2021 Q1 under time pressure.

4 Module map

Subtopic	**Why it matters**	**Exam appearances / practice**	**Core technique**	**Common trap**
Strategic form	Foundation for every equilibrium definition.	June 2020 Q4 asks define strategic game and NE; Class19–20.	Write $G = (N, (S_i), (u_i))$ and quantify over s_i'.	Calling actions strategies in dynamic games.
Dominance and IDSDS	Exams ask weak dominance and deletion arguments.	August 2024 Q2; PS9.	Inequality for all opponent strategies; strict somewhere.	Checking only on-path strategies.
Pure NE	Fast points in matrix games and continuations.	June 2019 Q1; stage games in repeated problems.	Best-response table.	Forgetting ties are best responses.
Mixed NE	High-yield computation and proof skill.	June 2024 Q2; PS9 Battle of Sexes.	Support enumeration plus off-support inequalities.	Indifference only, no off-support check.

Subtopic	Why it matters	Exam appearances / practice	Core technique	Common trap
Existence	Needed to know when DFG/Nash applies.	Class21; PS9; August 2021 Q1(d).	Compact/convex strategy sets, continuity, quasiconcavity; Kakutani.	Invoking DFG in infinite-player OLG.
Supermodular games	Useful for strategic complementarities and monotone equilibria.	SPMGames22; differentiated Bertrand.	Increasing best replies; Tarski fixed point.	Assuming uniqueness from supermodularity.
Extensive-form definitions	Strategy-set questions are frequent.	Class23–24; June 2023 Q1; August 2024 Q2.	Histories, information sets, actions, terminal histories.	Strategy for later mover is not a single action.
Normal-form representation	Converts trees into NE questions.	dgt02; August 2024 Q2(a).	Strategy profiles induce terminal histories and expected payoffs.	Omitting off-path components.
SPNE	Main refinement for dynamic complete-information games.	June 2023 Q1; August 2024 Q2; PS10.	NE in every subgame; backward induction.	Verifying only the reached subgame.
Sequential equilibrium	Main refinement for imperfect-information trees.	June 2023 Q2; June 2018 Q4.	Assessment (σ, μ); Bayes on path; perturbation off path.	Free off-path beliefs that are inconsistent with limits.
Finite repetition	Identifies unraveling vs reward/punishment.	Class25; June 2019 Q1; PS11.	Last-period NE; induction.	Saying all finite repeated games unravel.
Infinite repetition	Folk theorem and collusion.	Class25; RepeatedGames FT; PS11.	One-shot deviation IC; Nash reversion.	Ignoring profitable one-time deviations.
OLG repeated games	Infinite horizon with finite-lived players.	August 2021 Q1; August 2013 Q2.	Write player-specific strategy functions; young IC.	Treating it as a two-player infinitely repeated game.

5 Learning objectives

By the end of this module, the student can:

1. Given a strategic-form game $G = (N, (S_i), (u_i))$, write the definitions of dominant strategy equilibrium, strictly/weakly dominated strategy, IDSDS, Nash equilibrium, mixed extension, and support condition.
2. Given a payoff matrix, find all pure Nash equilibria and all mixed Nash equilibria by support

enumeration with off-support checks.

3. Given a continuous game, check the Debreu–Fan–Glicksberg assumptions and draw/intersect best replies.
4. Given an extensive-form game, write every player's pure strategy set as a set of functions from information sets to actions.
5. Given a Stackelberg-Cournot game, derive the follower's entire response function, leader's objective, SPNE, and a Nash equilibrium that is not SPNE.
6. Given an imperfect-information tree, compute receiver belief thresholds, write an assessment, and verify sequential rationality and consistency.
7. Given a finitely repeated stage game, determine whether backward induction pins down all SPNE outcomes or whether multiple stage NE permit early non-stage play.
8. Given an infinitely repeated game, write the trigger strategy and incentive constraint that supports a target path.
9. Given an OLG game, write each finite-lived player's strategy set and distinguish Nash from subgame-perfect verification.

6 Strategic-form games

6.1 Definitions

Definition 6.1 (Strategic-form game)**.** *A strategic-form game is a tuple*

$$G = \big(N, (S_i)_{i\in N}, (u_i)_{i\in N}\big),$$

where $N = \{1, \dots, n\}$ is the set of players, S_i is player i's strategy set, $S = \prod_i S_i$, and $u_i : S \to \mathbb{R}$ represents player i's preferences over strategy profiles.

Definition 6.2 (Best response and Nash equilibrium)**.** *For $s_{-i} \in S_{-i}$,*

$$\mathrm{BR}_i(s_{-i}) = \arg\max_{s_i \in S_i} u_i(s_i, s_{-i}).$$

A strategy profile $s^* \in S$ is a Nash equilibrium if, for every $i \in N$,

$$u_i(s_i^*, s_{-i}^*) \geq u_i(s_i, s_{-i}^*) \quad \textit{for every } s_i \in S_i.$$

Equivalently, $s_i^* \in \mathrm{BR}_i(s_{-i}^*)$ for all i.

Definition 6.3 (Dominance)**.** *A strategy s_i strictly dominates $\hat{s}_i$ if*

$$u_i(s_i, s_{-i}) > u_i(\hat{s}_i, s_{-i}) \quad \forall s_{-i} \in S_{-i}.$$

It weakly dominates $\hat{s}_i$ if the inequality is weak for all s_{-i} and strict for at least one s_{-i}. A strategy is strictly dominated if some other strategy strictly dominates it. A dominant strategy is one that weakly or strictly dominates all alternatives, depending on the convention stated in the problem.

Qualifier warning. The quantifier is over *all* opponent strategies, not only strategies that are plausible, on path, or undominated. Weak dominance is especially dangerous in extensive-form games because off-path components matter for the normal-form payoff comparison.

Proposition 6.4 (NE survive iterative deletion of strictly dominated strategies)**.** *In a finite strategic-form game, if s^* is a Nash equilibrium, then every component s_i^* survives iterative deletion of strictly dominated strategies.*

Proof script. At any deletion round, suppose s_i^* were strictly dominated in the reduced game by $\hat{s}_i$. Since s_{-i}^* has survived previous rounds, strict dominance gives

$$u_i(\hat{s}_i, s_{-i}^*) > u_i(s_i^*, s_{-i}^*),$$

contradicting the Nash inequality. Induct over deletion rounds.

6.2 Mixed extension and support conditions

For a finite game, let $\Delta(S_i)$ be the set of probability distributions over S_i. A mixed strategy profile is $\sigma = (\sigma_i)_i \in \prod_i \Delta(S_i)$. Expected utility is

$$U_i(\sigma) = \sum_{s \in S} u_i(s) \prod_{j \in N} \sigma_j(s_j).$$

Proposition 6.5 (Support test for mixed Nash equilibrium)**.** *A mixed profile σ^* is a Nash equilibrium iff, for every player i,*

$$\sigma_i^*(s_i) > 0 \quad \Longrightarrow \quad s_i \in \arg\max_{\hat{s}_i \in S_i} U_i(\hat{s}_i, \sigma_{-i}^*).$$

Equivalently, all pure strategies used with positive probability yield the same maximal payoff, and all unused pure strategies yield no more.

Qualifier warning. Support indifference is necessary but not sufficient unless you also check unused strategies. Many ASU mixed-equilibrium questions are designed to punish this omission.

6.3 Mechanical method: finite matrix games

1. **Pure NE.** Underline each player's best responses. A cell is a pure NE iff all players' payoffs are underlined.
2. **Mixed NE.** Choose candidate supports $T_i \subseteq S_i$. Make every player indifferent among strategies in T_i. Solve for probabilities. Then verify: probabilities are in $[0, 1]$, sum to one, and all strategies outside T_i are weakly worse.
3. **Dominance.** Check dominance inequalities against every opponent pure strategy; for mixed dominance, solve the linear inequalities.

6.4 Closed form: 2×2 mixed equilibrium

Consider

	L	R
U	(a, b)	(c, d)
D	(e, f)	(g, h)

where the row player mixes p on U and the column player mixes q on L. If both players mix with full support, then

$$q = \frac{g - c}{a - c - e + g}, \qquad p = \frac{h - f}{b - d - f + h},$$

provided denominators are nonzero and $p, q \in (0, 1)$. If either probability is outside $(0, 1)$, the full-support candidate fails; check boundary supports.

6.5 Example: Battle of the Sexes with $b > a > 0$

	W	O
W	(b, a)	$(0, 0)$
O	$(0, 0)$	(a, b)

There are two pure NE: (W, W) and (O, O). In the mixed NE, row is indifferent when column chooses W with probability q:

$$qb = (1 - q)a \quad \Rightarrow \quad q = \frac{a}{a + b}.$$

Column is indifferent when row chooses W with probability p:

$$pa = (1 - p)b \quad \Rightarrow \quad p = \frac{b}{a + b}.$$

Thus row plays W with probability $b/(a + b)$ and column plays W with probability $a/(a + b)$.

6.6 Example: June 2024 parametric 3×3 game

Rows are A, B, C; columns are D, E, F:

	D	E	F
A	$(0, 0)$	$(1, \beta)$	$(\beta, 1)$
B	$(\beta, 1)$	$(0, 0)$	$(1, \beta)$
C	$(1, \beta)$	$(\beta, 1)$	$(0, 0)$

Let row mix $p = (p_A, p_B, p_C)$ and column mix $q = (q_D, q_E, q_F)$. Row payoffs are

$$U_A = q_E + \beta q_F, \quad U_B = \beta q_D + q_F, \quad U_C = q_D + \beta q_E.$$

Column payoffs are

$$V_D = p_B + \beta p_C, \quad V_E = \beta p_A + p_C, \quad V_F = p_A + \beta p_B.$$

For $\beta \neq 1$ close to one, the unique fully mixed candidate is uniform:

$$p_A = p_B = p_C = q_D = q_E = q_F = \frac{1}{3}.$$

The support-enumeration proof shows no smaller support survives when β is sufficiently close to but not equal to one. At $\beta = 1$, the best-response inequalities collapse. The full set of mixed NE is:

$$\operatorname{supp}(p) \subseteq \{A : q_D = m_q\} \cup \{B : q_E = m_q\} \cup \{C : q_F = m_q\},$$
$$\operatorname{supp}(q) \subseteq \{D : p_A = m_p\} \cup \{E : p_B = m_p\} \cup \{F : p_C = m_p\},$$

where $m_q = \min\{q_D, q_E, q_F\}$ and $m_p = \min\{p_A, p_B, p_C\}$. This is a compact way to state all equilibria: a player puts positive probability only on the action corresponding to an opponent probability that is minimal.

Qualifier warning. Continuity of equilibrium correspondences is a set-valued question. Here the unique equilibria for $\beta \neq 1$ converge to the uniform equilibrium, but at $\beta = 1$ there are additional equilibria. Thus the graph can be closed while lower hemicontinuity fails.

7 Equilibrium existence theorems

Theorem 7.1 (Debreu–Fan–Glicksberg). *Let $G = (N, (S_i), (u_i))$ with finite N. If for each i:*

(a) S_i is nonempty, compact, and convex in a Euclidean space;

(b) u_i is continuous on S;

(c) $u_i(\cdot, s_{-i})$ is quasiconcave on S_i for every s_{-i},

then G has a Nash equilibrium.

Proof script. Let $\mathrm{BR}_i(s_{-i}) = \arg\max_{s_i \in S_i} u_i(s_i, s_{-i})$. Compactness and continuity give nonempty best replies. Quasiconcavity gives convex-valued best replies. The Maximum Theorem gives upper hemicontinuity. Define $\mathrm{BR}(s) = \prod_i \mathrm{BR}_i(s_{-i})$. Since $S = \prod_i S_i$ is nonempty, compact, and convex, Kakutani yields $s^* \in \mathrm{BR}(s^*)$, which is exactly NE.

Theorem 7.2 (Nash's finite-game mixed existence theorem). *Every finite strategic-form game has a mixed-strategy Nash equilibrium.*

Proof script. The mixed strategy set $\Delta(S_i)$ is a nonempty, compact, convex simplex. Expected utility is continuous and affine, hence quasiconcave, in own mixed strategy. Apply Debreu–Fan–Glicksberg to the mixed extension.

7.1 When not to invoke DFG

DFG does not apply if there are infinitely many players, if strategy sets are not compact subsets of finite-dimensional Euclidean space, if strategy sets are not convex, or if payoffs are not continuous/quasiconcave in own strategies. It also does not characterize equilibrium. It only gives existence.

Example 7.3 (All-pay/product-development race)**.** *If two firms choose expenditures in* $[0,1]$ *and the higher spender wins a prize, the payoff discontinuity at ties breaks continuity. Pure equilibrium may fail, but a mixed equilibrium may exist by direct construction if one works with distributions. Do not cite Nash's finite-game theorem when the pure strategy set is a continuum.*

7.2 Supermodular games and Tarski

Some ASU materials include games with strategic complementarities. The key theorem is not DFG but Tarski/Topkis.

Definition 7.4 (Increasing differences)**.** *A function* $u_i(a_i, a_{-i}, t)$ *has increasing differences in* a_i *and* (a_{-i}, t) *if for* $a_i' > a_i$ *and* $(a_{-i}', t') \geq (a_{-i}, t)$,

$$u_i(a_i', a_{-i}', t') - u_i(a_i, a_{-i}', t') \geq u_i(a_i', a_{-i}, t) - u_i(a_i, a_{-i}, t).$$

Strict increasing differences replaces $\geq$ *by* $>$ *when both changes are strict.*

Theorem 7.5 (Topkis/Tarski existence of extremal equilibria)**.** *If each strategy set is a nonempty compact sublattice of* $\mathbb{R}$*, payoffs are continuous in own action, and best replies are increasing by increasing differences, then the game has a smallest and largest Nash equilibrium. The extremal equilibria are increasing in parameters that enter payoffs with increasing differences.*

> **Qualifier warning.** Supermodularity gives existence of extremal equilibria and monotone comparative statics, not uniqueness. Multiple equilibria are often the point.

8 Static economic games

8.1 Hotelling location game

Two vendors choose $s_i \in [0,1]$. Consumers are uniformly distributed on the unit interval and buy from the closer vendor; ties split the market. A vendor wants to maximize market share. The unique NE is

$$s_1^* = s_2^* = \frac{1}{2}.$$

Proof: if $s_1 < s_2$, each player wants to move closer to the other to capture more market. Hence any equilibrium must have $s_1 = s_2$. If the common location differs from $1/2$, a vendor can move slightly toward the longer side and capture more than half the market. At $1/2$, neither can get more than half by moving.

8.2 Cournot oligopoly: linear demand

Let $P(Q) = a - Q$ for relevant Q, constant unit cost $k < a$, and $q_i \geq 0$. Firm i's profit is

$$\pi_i(q_i, q_{-i}) = (a - q_i - Q_{-i} - k)q_i.$$

The best response is

$$R_i(Q_{-i}) = \max\left\{0, \frac{a - k - Q_{-i}}{2}\right\}.$$

In a symmetric interior equilibrium with n firms,

$$q_i^* = \frac{a-k}{n+1}, \qquad Q^* = \frac{n(a-k)}{n+1}, \qquad P^* = \frac{a+nk}{n+1}.$$

As $n \to \infty$, $P^* \to k$.

Four-layer answer. For Cournot problems:

1. Object: actions $q_i \geq 0$, payoff $\pi_i = (P(Q) - k_i)q_i$.
2. Feasibility: impose nonnegativity and any capacity/compactness bound.
3. Optimality: compute $R_i(Q_{-i})$ with corner checks.
4. Verification: solve intersection and verify no corner omitted.

8.3 Bertrand with identical goods and constant marginal cost

With two firms, identical products, constant unit cost k, and textbook tie-breaking, the unique pure-strategy NE is $p_1 = p_2 = k$. If $p_i > k$ and $p_j \geq p_i$, firm j can slightly undercut and capture the market. If $p_i < k$, firm i loses money on sales. At $p_i = p_j = k$, neither can profitably undercut or raise price.

Qualifier warning. The Bertrand conclusion is fragile. Product differentiation, capacity constraints, increasing marginal cost, or repeated interaction can overturn marginal-cost pricing.

8.4 Advertising game

In the ASU advertising problem,

$$\pi_i(x_i, x_j) = x_i(2 - x_i) - x_j.$$

If $x_i \in \mathbb{R}_+$ and no additional constraint binds, firm i's payoff is strictly concave in x_i with first-order condition $2 - 2x_i = 0$. Hence $x_i^* = 1$ for both firms and equilibrium profits are 0.

The Pareto problem for symmetric outcomes sets $x_1 = x_2 = x$ and each profit equals

$$\pi_i = x(2 - x) - x = x - x^2.$$

The symmetric Pareto optimum is $x = 1/2$, yielding $1/4$ for each, illustrating a prisoners'-dilemma structure: individually optimal advertising is excessive.

9 Extensive-form games

9.1 Game trees and information sets

A finite extensive-form game consists of players, a rooted tree of histories/nodes, a player or chance assignment at each nonterminal node, feasible actions at each decision node, information sets for each player, probabilities at chance nodes, and payoffs at terminal histories.

In the Kuhn-style notation used in the slides, an extensive form is

$$\Gamma = [I, (X, <), (P_i)_{i=0}^{I}, (\mathcal{U}_i)_{i=1}^{I}, (C_h)_{h\in\mathcal{U}_i}, (P_x)_{x\in P_0}, V],$$

where 0 is chance, $(X, <)$ is the tree, Z is the set of terminal nodes, P_i is the set of nodes where player i moves, $\mathcal{U}_i$ partitions P_i into information sets, C_h is the set of choices at information set h, P_x is the chance distribution, and $V : Z \to \mathbb{R}^I$ is the payoff function.

Definition 9.1 (Pure strategy in an extensive-form game)**.** *A pure strategy for player i is a function*

$$s_i : \mathcal{U}_i \to \bigcup_{h\in\mathcal{U}_i} C_h$$

such that $s_i(h) \in C_h$ for every information set $h \in \mathcal{U}_i$.

Definition 9.2 (Mixed and behavior strategies)**.** *A mixed strategy is a distribution over pure strategies, $\mu_i \in \Delta(S_i)$. A behavior strategy is a function assigning to each information set h a distribution over available actions, $\sigma_i(h) \in \Delta(C_h)$. In finite games with perfect recall, Kuhn's theorem says mixed and behavior strategies are outcome-equivalent in the relevant strategic sense.*

Definition 9.3 (Perfect recall and perfect information)**.** *A player has perfect recall if she remembers her previous information and actions. A game has perfect information if every information set is a singleton. If at least one information set is non-singleton, the game has imperfect information.*

9.2 Complete contingent plans: the exam's favorite definition trap

Consider a two-stage Stackelberg structure: player 1 chooses $a_1 \in A_1$; player 2 observes a_1 and chooses $a_2 \in A_2$; payoffs are $U_i(a_1, a_2)$. Then

$$S_1 = A_1,$$

but

$$S_2 = \{s_2 : A_1 \to A_2\}.$$

Player 2's strategy specifies an action after every possible a_1, not just after the equilibrium a_1.

Four-layer answer. For any extensive-form strategy-set question:

1. List each information set of player i.
2. List the feasible actions at each information set.
3. Write S_i as the product of feasible action sets across information sets, or as a function space.
4. Give one concrete example of a strategy.

9.3 Normal-form representation of an extensive-form game

Given pure strategies $s = (s_i)_i$, chance moves and strategies induce a distribution over terminal histories. The normal-form payoff is expected terminal payoff:

$$u_i(s) = \sum_{z\in Z} V_i(z) \Pr(z \mid s).$$

In the dgt02 example, player 2 observes a type/history and has strategies like RR, RL, LR, LL because a strategy is a function specifying R or L after each possible type/history. The normal-form payoff entries are expected values using chance probabilities.

Qualifier warning. Never write player 2's strategy as just R or L when player 2 has multiple information sets. Write RR, RL, LR, or LL, or write the function explicitly.

9.4 Subgames and SPNE

Definition 9.4 (Subgame)**.** *A subgame begins at a singleton information set (a node known to the mover), contains all successor nodes, and does not cut any information set. In perfect-information games, every decision node starts a subgame.*

Definition 9.5 (Subgame-perfect Nash equilibrium)**.** *A strategy profile s^* is a subgame-perfect Nash equilibrium if it induces a Nash equilibrium in every subgame.*

Theorem 9.6 (Backward induction)**.** *Every finite perfect-information game has a pure-strategy SPNE. It can be found by solving terminal subgames first and moving backward.*

Proof script. At each last decision node, choose an action maximizing the mover's payoff. Replace that node by its induced payoff. Continue backward. At every subgame, the chosen action is optimal given continuation strategies, so the resulting profile is SPNE.

Theorem 9.7 (One-shot deviation principle, finite version)**.** *In a finite extensive-form game with perfect recall, a strategy profile is SPNE if no player can profitably deviate at a single information set in any subgame, holding the rest of the strategy fixed.*

10 Canonical derivation: Stackelberg-Cournot

10.1 Environment

Two firms have constant unit cost k, inverse demand

$$\tilde{p}(Q) = \max\{0, a - Q\}, \qquad a > k > 0.$$

Firm 1 first chooses $q_1 \geq 0$. Firm 2 observes q_1 and chooses $q_2 \geq 0$. Payoffs are profits

$$U_i(q_1, q_2) = \tilde{p}(q_1 + q_2)q_i - kq_i.$$

10.2 Object and strategy sets

$$S_1 = \mathbb{R}_+, \qquad S_2 = \{s_2 : \mathbb{R}_+ \to \mathbb{R}_+\}.$$

For $s = (q_1, s_2)$, normal-form utility is

$$u_i(q_1, s_2) = U_i(q_1, s_2(q_1)).$$

A Nash equilibrium is (q_1^*, s_2^*) such that

$$u_1(q_1^*, s_2^*) \geq u_1(q_1, s_2^*) \quad \forall q_1 \geq 0,$$

$$u_2(q_1^*, s_2^*) \geq u_2(q_1^*, s_2) \quad \forall s_2 : \mathbb{R}_+ \to \mathbb{R}_+.$$

10.3 Follower best response

Given q_1, firm 2 solves

$$\max_{q_2 \geq 0} (a - q_1 - q_2 - k) q_2$$

when $q_1 + q_2 \leq a$; producing beyond the choke quantity cannot improve profits. The concave objective has FOC

$$a - k - q_1 - 2q_2 = 0.$$

Thus

$$R_2(q_1) = \max \left\{ 0, \frac{a - k - q_1}{2} \right\}.$$

10.4 Leader problem and SPNE

For $q_1 \leq a - k$, firm 1 anticipates $R_2(q_1)$ and earns

$$\pi_1(q_1, R_2(q_1)) = q_1 \frac{a - k - q_1}{2}.$$

This is maximized at

$$q_1^S = \frac{a-k}{2}, \qquad q_2^S = R_2(q_1^S) = \frac{a-k}{4}.$$

Profits are

$$\pi_1^S = \frac{(a-k)^2}{8}, \qquad \pi_2^S = \frac{(a-k)^2}{16}.$$

A SPNE is

$$s_1^* = \frac{a-k}{2}, \qquad s_2^*(q_1) = R_2(q_1) \quad \forall q_1 \geq 0.$$

10.5 A Nash equilibrium that is not SPNE

Define

$$s_1 = 0, \qquad s_2(q_1) = \begin{cases} (a-k)/2, & q_1 = 0, \\ a, & q_1 > 0. \end{cases}$$

On path, firm 2 best responds to $q_1 = 0$ by producing monopoly output $(a - k)/2$. Firm 1 earns 0. Any deviation $q_1' > 0$ induces $s_2(q_1') = a$, price 0, and firm 1 payoff $-kq_1' < 0$. Hence firm 1 does not deviate. Firm 2's on-path action is optimal. This is a Nash equilibrium.

It is not SPNE because after many off-path histories, such as $q_1' = (a - k)/2$, firm 2's prescribed action a is not a best response; the best response is $(a - k - q_1')/2 = (a - k)/4$.

10.6 Weak dominance in the Stackelberg game

The pointwise best-response strategy $s_2^B(q_1) = R_2(q_1)$ weakly dominates every other firm-2 strategy: for each q_1, it maximizes firm 2's payoff after that q_1. But it need not be used in every Nash equilibrium because normal-form Nash equilibrium only tests firm 2's action after the equilibrium q_1; off-path components can be nonoptimal and still not affect firm 2's equilibrium payoff.

11 Canonical derivation: owner-manager two-period game

11.1 Environment

There is an owner O and two managers M_1, M_2. At each date $t = 1, 2$, the owner chooses one manager to run the firm. The chosen manager i chooses expropriation $x_i^t \in [0,1]$. Current payoffs are

$$O : 1 - x_i^t, \qquad M_i : x_i^t, \qquad M_j : 0.$$

All players discount date 2 by $\delta \in (0,1)$.

11.2 Object: strategy sets

Owner strategy:

- date 1: choose M_1 or M_2;
- date 2: after every date-1 history (i, x_i^1), choose M_1 or M_2.

Thus

$$S_O = \{M_1, M_2\} \times \{f : [\{M_1, M_2\} \times [0,1]] \to \{M_1, M_2\}\}.$$

Manager i's strategy specifies an expropriation whenever chosen at date 1 and whenever chosen after every date-1 history at date 2.

11.3 Date-2 subgames

At date 2, the chosen manager maximizes $x \in [0,1]$, so chooses $x = 1$. The owner then receives 0 regardless of which manager is chosen. Hence the owner is indifferent between managers at date 2, and any date-2 hiring choice is sequentially rational as long as the chosen manager expropriates 1.

11.4 Highest SPNE owner payoff

Use date-2 hiring as a reward/punishment device. Suppose M_1 is chosen at date 1 and is asked to choose

$$x^* = 1 - \delta.$$

If M_1 chooses $x \le x^*$, hire M_1 at date 2; otherwise hire M_2. At date 2, whoever is hired expropriates 1.

Manager M_1's payoff from obeying is

$$x^* + \delta = 1.$$

The best deviation is to choose $x = 1$ and lose date-2 hiring, yielding 1. Thus obeying is optimal. The owner receives

$$1 - x^* = \delta$$

at date 1 and zero at date 2. So the owner gets δ.

No SPNE can give the owner more. If a date-1 manager chooses x, the most the owner can promise as a future reward to that manager is date-2 payoff 1. The manager can always deviate to $x = 1$ and get current payoff 1. Incentive compatibility requires

$$x + \delta \geq 1,$$

so $x \geq 1 - \delta$, and the owner's date-1 payoff $1 - x \leq \delta$.

11.5 Lowest SPNE owner payoff

The owner can get 0: choose either manager at date 1; every chosen manager expropriates 1 at every date; date-2 behavior is as above. The owner cannot get less than zero because $x \in [0, 1]$ implies $1 - x \geq 0$ each date.

11.6 A Nash equilibrium that is not SPNE

Take any SPNE path, and alter a manager's expropriation at an unreached date-2 subgame to 0. Since that subgame is never reached under the equilibrium path, no player's ex ante payoff changes and the profile remains a Nash equilibrium. It is not SPNE because if the subgame were reached, the chosen manager would profitably deviate from 0 to 1.

> **Qualifier warning.** For NE-not-SPNE examples, you do not need to change on-path behavior. A noncredible off-path continuation in an unreached subgame is enough, provided no one can profitably reach it by a unilateral deviation that improves payoff.

12 Sequential equilibrium

12.1 Definition

Definition 12.1 (Assessment)**.** *An assessment is a pair (σ, μ) where σ is a behavior-strategy profile and μ assigns to every information set a probability distribution over nodes in that information set.*

Definition 12.2 (Sequential equilibrium)**.** *An assessment (σ, μ) is a sequential equilibrium if:*

(i) ***Sequential rationality:*** *at every information set, the continuation strategy maximizes the moving player's expected continuation payoff given beliefs μ and other players' strategies;*

(ii) ***Consistency:*** *there exists a sequence of completely mixed behavior profiles $\sigma^n \to \sigma$ such that the beliefs induced by Bayes' rule from σ^n converge to μ.*

> **Qualifier warning.** Weak perfect Bayesian equilibrium lets off-path beliefs be almost arbitrary. Sequential equilibrium does not: off-path beliefs must be limits of Bayes-rule beliefs under trembles.

12.2 Belief threshold method

At a non-singleton information set, let α be the belief of being at the top node. Compute expected payoffs for each receiver action as functions of α. Partition $[0, 1]$ into regions where each action is optimal. Then impose sender incentives and Bayes consistency.

12.3 Worked example: June 2023 sequential-equilibrium tree

In the June 2023 tree, nature chooses H or T with probability 1/2. Player 1 observes the state. Action x_z enters player 2's information set; action y_z ends the game at $(0, 0)$. Player 2's information set has a top node a and a bottom node b, with actions e, f, g and payoffs:

	e	f	g
top a	$(3, 8)$	$(-1, 7)$	$(2, 6)$
bottom b	$(1, 0)$	$(-2, 7)$	$(-1, 9)$

Let α be player 2's belief of being at the top node. Player 2's expected payoffs are

$$U_2(e \mid \alpha) = 8\alpha, \qquad U_2(f \mid \alpha) = 7, \qquad U_2(g \mid \alpha) = 9 - 3\alpha.$$

Therefore

$$\mathrm{BR}_2(\alpha) = \begin{cases} \{g\}, & \alpha < 2/3, \\ \{f, g\}, & \alpha = 2/3, \\ \{f\}, & 2/3 < \alpha < 7/8, \\ \{e, f\}, & \alpha = 7/8, \\ \{e\}, & \alpha > 7/8. \end{cases}$$

If player 2 mixes $q = (q_e, q_f, q_g)$, player 1's payoff from entering is

$$\text{top: } 3q_e - q_f + 2q_g, \qquad \text{bottom: } q_e - 2q_f - q_g.$$

Exiting gives 0.

Two useful classes of sequential equilibria follow.

No-entry equilibria. Both types of player 1 choose y. The information set is off path. Consistency permits any limiting belief $\alpha \in [0, 1]$, but sequential rationality and no-entry incentives restrict (α, q):

- if $\alpha = 2/3$, player 2 may mix between f and g with $q_f \geq 2/3$;
- if $2/3 < \alpha < 7/8$, player 2 plays f;
- if $\alpha = 7/8$, player 2 may mix between e and f with $q_e \leq 1/4$.

A mixed-entry equilibrium. The top type enters for sure. The bottom type enters with probability 1/7. Player 2's belief is then

$$\alpha = \frac{1}{1 + 1/7} = \frac{7}{8}.$$

At $\alpha = 7/8$, player 2 may mix between e and f. Choose

$$q_e = \frac{2}{3}, \qquad q_f = \frac{1}{3}.$$

The bottom type's entry payoff is

$$1 \cdot \frac{2}{3} - 2 \cdot \frac{1}{3} = 0,$$

so it is willing to mix. The top type's entry payoff is

$$3 \cdot \frac{2}{3} - 1 \cdot \frac{1}{3} = \frac{5}{3} > 0,$$

so it enters for sure. This verifies sequential rationality and Bayes consistency.

13 Graphical tools

13.1 Payoff matrix underlining

For each column, underline row payoffs that are maximal; for each row, underline column payoffs that are maximal. Cells with both underlines are pure NE. Mark ties carefully.

13.2 Best-response diagrams

For two-player continuous games with $s_i \in [a_i, b_i]$, plot $R_1(s_2)$ and $R_2(s_1)$ in the same square. Intersections are pure NE. If best replies are correspondences, use vertical or horizontal segments. If the game is supermodular, increasing best replies make extremal intersections especially important.

13.3 Extensive-form trees

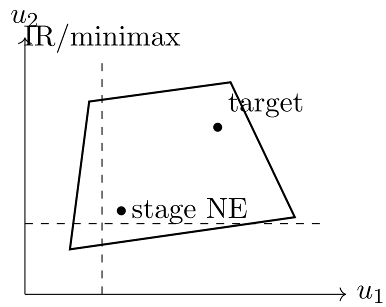
Use three visual conventions:

- decision nodes labelled by the mover;
- dashed ovals/boxes for information sets;
- terminal payoffs written in player order.

A subgame can start only at a singleton information set and cannot cut an information set. Draw a box around the candidate subgame before applying backward induction.

13.4 Repeated-game payoff diagrams

For two-player repeated games, draw the feasible payoff set, the minimax/payoff floor, and the target payoff. A trigger-strategy proof asks whether the target payoff is high enough relative to the one-shot deviation payoff and the punishment payoff.



14 Repeated games

14.1 Definitions

Let $G = (N, (A_i), (\pi_i))$ be a simultaneous-move stage game and $A = \prod_i A_i$. A repeated game with horizon $T \in \mathbb{N} \cup \{\infty\}$ has histories

$$H_t = \{h^0 a^1 \cdots a^t : a^\tau \in A,\ \tau = 1, \ldots, t\}.$$

At date t, each player observes h^{t-1} and chooses an action $a_i^t \in A_i$. A pure strategy is a sequence

$$s_i = (f_i^1, f_i^2, \ldots), \qquad f_i^t : H_{t-1} \to A_i.$$

For finite T,

$$U_i(h^T) = \sum_{t=1}^{T} \delta^{t-1} \pi_i(a^t).$$

For $T = \infty$, require $\delta \in (0, 1)$ and use

$$U_i(h^\infty) = \sum_{t=1}^{\infty} \delta^{t-1} \pi_i(a^t),$$

or the normalized version $(1-\delta) \sum_t \delta^{t-1} \pi_i(a^t)$ if the problem uses average discounted payoffs.

14.2 Finite repetition with unique stage NE

Proposition 14.1. *If the stage game has a unique Nash equilibrium a^*, then every finite repetition has a unique SPNE outcome: a^* is played after every history at every date.*

> **Proof script.** At the final date, every subgame is the stage game, so a^* must be played. Move one date backward. Since future play is fixed at a^* regardless of current actions, current incentives are exactly the stage-game incentives, so a^* must be played. Induct backward.

For the prisoners' dilemma

	C	D
C	$(2, 2)$	$(-1, 3)$
D	$(3, -1)$	$(0, 0)$

the unique stage NE is (D, D), so the unique finite-horizon SPNE is: choose D at every history and every date.

Qualifier warning. The strategy "defect every period after every history" is not necessarily a dominant strategy in the finitely repeated game. It is the unique SPNE strategy profile in the finitely repeated prisoners' dilemma, but dominance quantifies over all opponent strategies. If the opponent conditions future cooperation on current cooperation, current cooperation may be better than current defection.

14.3 Finite repetition with multiple stage NE

If the stage game has multiple Nash equilibria, early non-stage-NE action profiles may be supportable by continuation rewards and punishments. The method is:

1. Identify all stage-game NE and their payoff vectors.
2. At the last date, continuation must be one of these NE payoff vectors.
3. At the previous date, a target action profile a is supportable if for each player i and each deviation a_i',
$$\pi_i(a) + V_i^{\text{reward}} \geq \pi_i(a_i', a_{-i}) + V_i^{\text{punish}}(a_i'),$$
where reward and punishment continuation payoffs come from stage-game NE.
4. Iterate backward for longer horizons.

This is the June 2019 Q1 logic. The stage game has two pure NE, (a_1, a_2) with payoff $(5, 3)$ and (b_1, b_2) with payoff $(2, 2)$. In a twice-repeated game, date 2 must be one of these; date 1 can sometimes be non-Nash if the continuation switches from $(5, 3)$ to $(2, 2)$ after deviations in a way that satisfies the inequalities.

14.4 Infinite repeated prisoners' dilemma and grim trigger

Grim trigger: play C in period 1; after any history, play C iff all previous play was (C, C); otherwise play D forever.

If both use grim trigger, cooperation gives
$$V_C = \frac{2}{1-\delta}.$$
A one-shot deviation gives 3 now and 0 forever:
$$V_D = 3.$$
Thus cooperation is optimal iff
$$\frac{2}{1-\delta} \geq 3 \quad \Longleftrightarrow \quad \delta \geq \frac{1}{3}.$$
After any history with a past deviation, both play D forever, which is a Nash equilibrium of the continuation repeated game. Therefore grim trigger is SPNE for $\delta \geq 1/3$.

14.5 Folk theorem with Nash reversion

Let $\tilde{a} \in \mathrm{NE}(G)$ be a stage-game Nash equilibrium and let a^* be a target action profile satisfying

$$\pi_i(a^*) > \pi_i(\tilde{a}) \quad \forall i.$$

The Nash-reversion strategy plays a^* until a deviation occurs, then plays $\tilde{a}$ forever. If $V_i(a^*_{-i}) = \sup_{a_i} \pi_i(a_i, a^*_{-i}) < \infty$, the no-deviation condition is

$$\frac{\pi_i(a^*)}{1-\delta} \geq V_i(a^*_{-i}) + \frac{\delta \pi_i(\tilde{a})}{1-\delta}.$$

Equivalently,

$$\delta \geq \frac{V_i(a^*_{-i}) - \pi_i(a^*)}{V_i(a^*_{-i}) - \pi_i(\tilde{a})}$$

for every player with a positive denominator. For sufficiently high δ, the target path is a SPNE outcome.

14.6 Repeated Bertrand collusion

With homogeneous Bertrand, stage NE is price k and profit 0. Let firms collude on monopoly price p^M and split demand, so each gets $\pi^M/2$ per period. A deviation slightly undercuts and captures nearly π^M once, followed by zero punishment. The IC is

$$\frac{\pi^M/2}{1-\delta} \geq \pi^M,$$

so

$$\delta \geq \frac{1}{2}.$$

This is the clean repeated-interaction answer to the class note that Bertrand's one-shot marginal-cost conclusion can be overturned by repeated interaction.

15 Overlapping-generations repeated games

15.1 Environment

At each date $t \geq 1$, player $t-1$ is old and player t is young. They play the prisoners' dilemma

	C (young)	D (young)
C (old)	$(2,2)$	$(-1,3)$
D (old)	$(3,-1)$	$(0,0)$

Player 0 lives only at date 1. Each player $i \geq 1$ lives at dates i and $i+1$, discounting the old-age payoff by $\delta \in (0,1)$.

15.2 Strategy sets

Let H^{t-1} be the set of public histories through date $t-1$. Player 0 has a one-date strategy

$$f_0^1 : H^0 \to \{C, D\}.$$

For $i \geq 1$, a pure strategy is a pair

$$s_i = (f_i^i, f_i^{i+1}),$$

where

$$f_i^i : H^{i-1} \to \{C, D\} \quad \text{(young-age action)},$$
$$f_i^{i+1} : H^i \to \{C, D\} \quad \text{(old-age action)}.$$

15.3 Supporting young cooperate, old defect

Target outcome: at each date, the old player chooses D and the young player chooses C.

Strategy profile:

- If no past deviation from the target outcome has occurred, a young player chooses C and an old player chooses D.
- After any deviation, all future players choose D whenever they move.

A young player who follows the strategy receives -1 when young and 3 when old:

$$U^{\text{follow}} = -1 + 3\delta.$$

A young player who deviates to D receives 0 now and triggers future punishment, giving old-age payoff 0:

$$U^{\text{deviate}} = 0.$$

Thus young cooperation is incentive compatible iff

$$-1 + 3\delta \geq 0 \quad \iff \quad \delta \geq \frac{1}{3}.$$

An old player on path prefers D to C against a young C because $3 > 2$, and old players have no future. After a punishment history, D is a stage-game best response to D. Therefore the strategy is a Nash equilibrium, and the same checks after every public history give SPNE when $\delta \geq 1/3$.

15.4 Why DFG is not the right existence theorem here

The OLG game has countably many players and each player's strategy is a function on a growing public history space. It is not a finite-player game with compact convex strategy sets in a finite-dimensional Euclidean space. Therefore Debreu–Fan–Glicksberg cannot be invoked directly to justify existence. Existence must be shown by explicitly constructing and verifying a strategy profile, as above.

16 Proof-writing templates

16.1 Template 1: prove a profile is Nash

Four-layer answer. Object. State S_i and u_i.
Feasibility. Confirm $s_i^* \in S_i$.
Optimality. For arbitrary $s_i \in S_i$, compute or bound $u_i(s_i^*, s_{-i}^*) - u_i(s_i, s_{-i}^*)$.
Verification. Repeat for every player. If mixed, check support and off-support inequalities.

16.2 Template 2: prove a profile is SPNE

Four-layer answer. Object. List subgames.
Feasibility. State full strategies, including off-path actions.
Optimality. In each subgame, show the restricted strategy profile is NE. Use backward induction if finite perfect information.
Verification. Mention every class of subgames; do not only verify the equilibrium path.

16.3 Template 3: prove not SPNE

Find a subgame and a player with a profitable deviation within that subgame. It is enough to identify one off-path subgame where the prescribed continuation is not Nash.

16.4 Template 4: sequential equilibrium verification

1. Write the assessment (σ, μ).
2. At each information set, compute continuation payoffs under μ.
3. Check sender/player optimality at every type or information set.
4. Check Bayes rule on path.
5. For off-path information sets, either construct a tremble sequence or state the limiting ratio that generates the belief.

16.5 Template 5: repeated-game incentive constraint

For a target path with punishment value P_i and deviation payoff D_i, write

$$\underbrace{\frac{R_i}{1-\delta}}_{\text{follow target}} \geq \underbrace{D_i + \frac{\delta P_i}{1-\delta}}_{\text{deviate once then punishment}}.$$

Then solve for δ and check punishment is credible.

17 Common traps

1. Writing a later mover's strategy as an action rather than a function from histories/information sets to actions.

2. Finding a path of play but not a full strategy profile.
3. Calling a profile SPNE after checking only the reached subgame.
4. In mixed NE, imposing indifference on the support but forgetting off-support inequalities.
5. In weak dominance, checking only equilibrium-path opponent strategies.
6. Treating weakly dominant strategies as if they must appear in every NE. They need not if dominance concerns off-path components of an extensive-form strategy.
7. Using DFG where strategy sets are finite but not convex without passing to the mixed extension.
8. Using Nash's finite-game theorem for continuum-action games.
9. Forgetting corner checks in Cournot and Stackelberg best responses.
10. Confusing WPBE with sequential equilibrium: off-path beliefs cannot always be chosen freely under sequential equilibrium.
11. Saying finite repeated games always unravel. They unravel cleanly when the stage game has a unique NE; multiple stage NE can create rewards and punishments.
12. Forgetting that OLG players are finite-lived, so each player has only two decision dates even though the game has infinitely many dates.

18 Exam variations

18.1 How ASU mutates the archetypes

Archetype	**Mutation**	**Invariant solution move**	**Example**
Strategy-set writing	Stackelberg outputs, owner-manager histories, OLG age-specific histories.	Strategy is a function from information sets/histories to actions.	June 2023 Q1; August 2024 Q2; August 2021 Q1.
Weak dominance	Follower has pointwise best response but NE may use off-path dominated components.	Quantify over all leader outputs; then build counterexample NE.	August 2024 Q2.
Mixed equilibrium	Parameter makes a continuum at knife-edge.	Indifference equations plus set-valued continuity language.	June 2024 Q2.
SPNE payoff bounds	Future hiring/reward disciplines current expropriation.	Write IC $x + \delta \geq 1$ and optimize owner payoff.	June 2023 Q1.
Sequential equilibrium	Nature move plus receiver information set.	Belief threshold, sender entry incentives, consistency by perturbation ratios.	June 2023 Q2; June 2018 Q4.

Archetype	Mutation	Invariant solution move	Example
Finite repetition	Stage game has one or multiple NE.	Last-period NE; reward/punishment continuation values.	June 2019 Q1; PS11.
OLG cooperation	Infinite chain of finite-lived players.	Young's sacrifice compared with old-age reward.	August 2021 Q1; August 2013 Q2.
Existence theorem check	Infinite players or discontinuities.	State exact theorem assumptions and show which fail.	August 2021 Q1(d); PS9.

19 Exercise bank

19.1 Mechanical drills

M1. **Strategy functions.** In a Stackelberg game with finite $A_1 = \{L, R\}$ and $A_2 = \{u, d\}$, write S_1 and S_2 and count strategies.

M2. **Normal form.** A chance move selects $t = a$ with probability .6 and $t = b$ with probability .4. Player 2 observes t and chooses R or L. Write player 2's pure strategies and the formula for expected payoff.

M3. **Pure NE.** Find all pure NE of the June 2019 stage game with payoffs $(5, 3), (2, 2)$ on diagonal entries $(a, a), (b, b)$ and zeros otherwise except $(a, c) = (2, 0)$.

M4. **Weak dominance.** State the inequality proving that one firm-2 Stackelberg strategy weakly dominates another.

M5. 2×2 **mixing.** Derive the mixed equilibrium of Battle of the Sexes with payoffs $b > a > 0$.

M6. **Subgames.** Give the formal reason an information set with two nodes cannot start a subgame.

M7. **Repeated histories.** For a two-action two-player stage game, write H_0, H_1, H_2 and a pure strategy component f_i^3.

M8. **Belief threshold.** In the June 2023 sequential tree, compute player 2's best response at beliefs $\alpha = 1/2, 2/3, 7/8, 1$.

M9. **OLG strategies.** Write the pure strategy of player $i \geq 1$ in the OLG prisoners' dilemma.

M10. **Off-support check.** Explain what must be checked after solving indifference equations for a mixed NE.

19.2 Proof and characterization drills

P1. Prove that every NE survives iterated deletion of strictly dominated strategies.

P2. Prove the support condition for finite mixed Nash equilibrium using affine programming.

P3. Give the proof skeleton of Debreu–Fan–Glicksberg.

P4. Prove finite backward induction gives a SPNE in perfect-information games.

P5. Prove that finite repeated prisoners' dilemma has unique SPNE outcome (D, D) every date.

P6. Derive the grim-trigger IC for the infinitely repeated prisoners' dilemma.

P7. Derive the Nash-reversion folk-theorem IC for a target action a^*.

P8. Show that the Stackelberg-Cournot follower's pointwise best-response strategy is weakly dominant.

P9. Prove the highest owner payoff in the owner-manager game is δ.

P10. Prove DFG cannot be invoked directly in the OLG game.

19.3 Past-qual applications

Q1. **June 2019 Q1.** Characterize pure stage NE and outline the SPE outcome algorithm for $\Gamma(2)$.

Q2. **June 2023 Q1.** Write the owner and manager strategy sets and compute highest/lowest SPNE owner payoff.

Q3. **June 2023 Q2.** Characterize sequential equilibria of the tree using belief thresholds.

Q4. **August 2024 Q2.** Derive the SPNE of Stackelberg-Cournot and construct a NE not SPNE.

Q5. **June 2024 Q2.** Solve the unique mixed NE for $\beta \neq 1$ close to one and state the equilibrium set at $\beta = 1$.

Q6. **August 2021 Q1.** Construct a NE/SPNE supporting young C, old D, and find the cutoff δ.

Q7. **June 2018 Q4.** Define sequential equilibrium in the three-person tree and identify why player 1 mixing can fail sequential rationality.

Q8. **June 2016 Q2.** In the parent-child game, write each player's strategy set and test the claim that every SPNE maximizes the child's action.

19.4 Trap / economist-claim questions

T1. "In a finitely repeated prisoners' dilemma, defect-after-every-history is a dominant strategy equilibrium." True or false?

T2. "If a player has a weakly dominant strategy, that strategy must be used in every Nash equilibrium." True or false?

T3. "Every Nash equilibrium of a perfect-information extensive-form game is subgame perfect." True or false?

T4. "To find a mixed equilibrium, it is enough to make players indifferent among the actions they use." True or false?

T5. "The Debreu–Fan–Glicksberg theorem proves the OLG game has a Nash equilibrium." True or false?

T6. "The folk theorem says every feasible payoff is supportable for every discount factor." True or false?

20 Answer sketches

20.1 Mechanical drills

M1. $S_1 = \{L, R\}$ and $S_2 = \{u, d\}^{\{L,R\}}$, so $S_2 = \{uu, ud, du, dd\}$ has four strategies. A strategy for player 2 specifies what to do after L and after R.

M2. Player 2's strategies are RR, RL, LR, LL. For any player i, $u_i(s) = .6V_i(z(a, s)) + .4V_i(z(b, s))$ where $z(t, s)$ is the terminal node induced by type t and strategies.

M3. Best responses give (a_1, a_2) and (b_1, b_2) as pure NE. No cell in row/column c is a NE because row a_1 or b_1 is better against the relevant column.

M4. s_2 weakly dominates $\hat{s}_2$ if $u_2(q_1, s_2) \geq u_2(q_1, \hat{s}_2)$ for every $q_1 \geq 0$, strict for some q_1. Pointwise best response satisfies this against all alternatives.

M5. Pure NE are $(W, W), (O, O)$. Mixed: row plays W with $b/(a+b)$; column plays W with $a/(a+b)$.

M6. A subgame must not cut information sets. If it starts at one node of a nonsingleton information set, it excludes another node the mover considers possible, changing the information structure.

M7. $H_0 = \{h^0\}$, $H_1 = \{h^0a^1 : a^1 \in A\}$, $H_2 = \{h^0a^1a^2 : a^1, a^2 \in A\}$. A date-3 component is $f_i^3 : H_2 \to A_i$.

M8. At $\alpha = 1/2$, g is best. At $2/3$, f and g tie. At $7/8$, e and f tie. At 1, e is best.

M9. $s_i = (f_i^i, f_i^{i+1})$ with $f_i^i : H^{i-1} \to \{C, D\}$ and $f_i^{i+1} : H^i \to \{C, D\}$.

M10. Check every unused pure strategy gives payoff no larger than the support payoff. Also verify probabilities are valid.

20.2 Proof and characterization drills

P1. Use contradiction at the first deletion round in which an equilibrium strategy is deleted. Strict dominance gives a profitable deviation against the surviving equilibrium opponents.

P2. Expected utility is affine in own mixed strategy. A mixture maximizes an affine function over a simplex iff every pure strategy in its support maximizes that affine function.

P3. Build best-reply correspondences. Use compactness/continuity for nonemptiness and upper hemicontinuity; quasiconcavity for convex values; apply Kakutani.

P4. Solve last decision nodes, replace by payoffs, and move backward. The constructed strategy is optimal in every subgame by induction.

P5. Last period must be (D, D). Given last τ periods are (D, D) after all histories, the previous period has the same incentives as the stage game, so it is (D, D); induct.

P6. Cooperation value $2/(1 - \delta)$; deviation value 3. IC gives $2/(1 - \delta) \geq 3$, so $\delta \geq 1/3$.

P7. Follow target: $\pi_i(a^*)/(1-\delta)$. Deviate: $V_i(a^*_{-i}) + \delta\pi_i(\tilde{a})/(1-\delta)$. Solve the inequality.

P8. For every q_1, $R_2(q_1)$ maximizes firm 2's payoff. Therefore the strategy $q_1 \mapsto R_2(q_1)$ gives at least as high payoff after every leader action and strictly higher when the alternative is not a best response.

P9. A manager can take 1 now if not sufficiently rewarded. IC requires $x + \delta \geq 1$, so $x \geq 1 - \delta$. Owner payoff $1 - x \leq \delta$, attainable with $x = 1 - \delta$.

P10. DFG is finite-player and finite-dimensional compact-convex. The OLG game has countably many players and strategy functions on infinite history spaces, so assumptions fail.

20.3 Past-qual applications

Q1. Stage NE are (a_1, a_2) and (b_1, b_2). In $\Gamma(2)$, date 2 must be one of these. Date 1 profiles are supportable if no one gains from deviating after accounting for date-2 reward/punishment NE payoffs.

Q2. S_O is first manager choice plus date-2 hiring rule after every date-1 history. Manager strategies specify expropriation whenever chosen. Highest SPNE owner payoff is δ; lowest is 0.

Q3. Compute $\mathrm{BR}_2(\alpha)$ thresholds 2/3 and 7/8. Combine with type-entry payoffs. No-entry SE use off-path beliefs in $[2/3, 7/8]$ with restrictions; mixed-entry SE has top enter, bottom enter with 1/7, and player 2 mix $(e, f) = (2/3, 1/3)$.

Q4. Follower $R_2(q_1) = \max\{0, (a - k - q_1)/2\}$. SPNE outputs are $((a-k)/2, (a-k)/4)$. A NE not SPNE: $q_1 = 0$, $s_2(0) = (a-k)/2$, and $s_2(q_1) = a$ for $q_1 > 0$.

Q5. For $\beta \neq 1$ close to one, unique mixed NE is uniform. At $\beta = 1$, use the minimum-probability support characterization in the text.

Q6. Strategy: young C, old D after clean histories; all D after deviations. Young IC is $-1 + 3\delta \geq 0$, so $\delta \geq 1/3$.

Q7. Define assessment (σ, μ). Check whether player 1's proposed mixing is optimal given continuation beliefs and strategies; often a positive probability on a strictly worse branch violates sequential rationality.

Q8. Child strategy is an action $a \in A$; parent strategy is a consumption allocation rule after each a. SPNE need not maximize the child's action unless parent's continuation choice and child's preferences make that action incentive compatible.

20.4 Trap questions

T1. False. It is SPNE in finitely repeated PD, but not dominant against opponents who condition future cooperation on current cooperation.

T2. False. Weak dominance need not pin down off-path components. The Stackelberg follower's pointwise best response is weakly dominant, but NE can use non-best off-path actions.

T3. False. Noncredible off-path threats can support Nash equilibria that fail in some subgame.

T4. False. Off-support actions must be weakly worse; otherwise the candidate is not an equilibrium.

T5. False. The OLG game violates DFG's finite-player/finite-dimensional compact-convex setup.

T6. False. Folk-theorem support needs patience and individual rationality; the simple Nash-reversion theorem also requires payoffs strictly above a stage-game NE payoff.

21 Past-qual practice map

Exam	**Question**	**Topic**	**Priority**
June 2013	Q1	Owner-manager two-period game; strategy sets; highest SPNE payoff; Nash not SPNE.	High
August 2013	Q2	OLG prisoners' dilemma; SPNE cooperation claim.	High
June 2013	Q4	Selten horse-style extensive form; subgames, SPNE, sequential equilibrium.	High
June 2016	Q2	Parent-child extensive form; strategies, SPNE claim, weak dominance.	Medium
June 2018	Q4	Three-person extensive-form game; all NE; sequential equilibrium definition.	Medium
June 2019	Q1	Static 3×3 NE; repeated game SPE outcomes for $T = 2, 3, 4$.	High
June 2019	Q2	Advertising game NE, Pareto optimum, core.	Medium
August 2021	Q1	OLG prisoners' dilemma; strategy sets; NE/SPNE support; DFG failure.	High
June 2023	Q1	Owner-manager game; full strategy sets; SPNE bounds; NE not SPNE.	High
June 2023	Q2	Sequential equilibrium in imperfect-information tree.	High
June 2024	Q2	Mixed NE and equilibrium-correspondence continuity.	High
August 2024	Q2	Stackelberg-Cournot; weak dominance; NE not SPNE; SPNE derivation.	High
PS9 2024	Q1–Q3	IDSDS proof, mixed Battle of Sexes, mixed equilibrium in continuum game.	High
PS10 2025	Q1–Q4	Extensive-form strategy sets, entry game, Nash vs SPNE.	High
PS11 2025	Q3–Q5	Repeated-game SPNE outcomes, finite PD dominance trap, repeated Bertrand.	High

22 Final study checklist

Before attempting old quals, confirm you can do the following without notes:

1. State $G = (N, (S_i), (u_i))$ and the Nash inequality from memory.
2. Define strict and weak dominance with the correct quantifiers.
3. Find all pure NE of a matrix in under two minutes.
4. Solve a 2×2 mixed equilibrium and check off-support actions.
5. State DFG and explain how Nash's finite-game theorem follows from it.
6. Write a later mover's extensive-form strategy as a function from histories/information sets to actions.
7. Define subgame and SPNE precisely.
8. Derive Stackelberg-Cournot outputs and a NE not SPNE.
9. Solve the owner-manager IC $x + \delta \geq 1$ and explain why the owner gets at most δ.
10. Define sequential equilibrium as an assessment satisfying sequential rationality and consistency.
11. Compute belief thresholds at a receiver information set.
12. Write repeated-game histories H_t and strategies $f_i^t : H_{t-1} \to A_i$.
13. Prove finite repeated PD unravels by induction.
14. Derive grim-trigger $\delta \geq 1/3$ for the ASU PD payoffs.
15. Write the OLG strategy set and derive the young-player IC $\delta \geq 1/3$.

23 Quality audit

- Source inventory included, with extraction limitations stated.
- Module map included with subtopic, relevance, exam anchors, technique, and trap.
- Canonical derivations use Object $\to$ Feasibility $\to$ Equilibrium/Optimality $\to$ Verification.
- Definitions emphasized: strategic form, dominance, NE, mixed extension, extensive form, strategy, subgame, SPNE, assessment, sequential equilibrium, repeated-game histories, OLG strategies.
- Equilibrium theorems included: Debreu–Fan–Glicksberg, Nash mixed existence, support condition, Tarski/Topkis, backward induction, one-shot deviation, finite repeated unravelling, Nash-reversion folk theorem.
- Graphical tools included: matrix underlining, best-response diagrams, tree/subgame diagrams, repeated-game payoff diagrams.
- Exercise bank includes mechanical, proof/characterization, past-qual, and trap/economist-claim questions with answer sketches.
- Past-qual practice map includes exact year/question references where identifiable.